

## CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 03 · LECTURE NOTES

# High Dynamic Range

*Measuring more range than one exposure can hold.*

## MODULE OUTCOMES

- Explain dynamic range and where a single exposure fails.
- Recover a camera response curve from bracketed exposures.
- Merge exposures into a radiance map with justified weights.
- Handle ghosting from moving subjects.
- Explain how modern cameras avoid bracketing entirely.

## 1 The dynamic range problem

|                                            | Dynamic range    | Note                                                                                                                            |
|--------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| A real scene — a sunlit room with a window | 20 stops or more | A ratio of a million to one between the brightest and darkest parts you care about. Outdoors with the sun in frame is far more. |
| Human eye, at a fixed adaptation level     | 10–14 stops      | <b>Much more over time</b> , because the eye adapts — which is part of why a scene looks fine to you and wrong in a photograph. |
| Camera sensor, single exposure             | 10–14 stops      | For a good modern sensor at base ISO. Considerably less at high ISO, because the noise floor rises.                             |
| 8-bit display image                        | ~8 stops         | <b>What you can actually show.</b> This is the constraint Module 04 exists to address.                                          |
| Print                                      | 6–7 stops        | Less still, and it depends on the paper and the viewing light.                                                                  |

**A stop is a factor of two in light.** Twenty stops is a ratio of about a million to one, and the display can show perhaps two hundred and fifty. **Compression is therefore not optional**, and the only question is how it is done — which is Module 04.

**A SINGLE EXPOSURE FORCES A CHOICE**

**Expose for the highlights and the shadows fall below the noise floor.** Expose for the shadows and the highlights clip.

**Clipped highlights are unrecoverable.** Every pixel in a clipped region reads the same saturation value, so the relative brightness information within it has been destroyed rather than merely compressed. No processing recovers it, and attempts to do so invent detail.

**Crushed shadows are nearly as bad** — the signal is present but buried in read noise, and lifting it amplifies the noise with it (Module 01).

**So: capture several exposures and combine them.** Each exposure measures a different portion of the scene's range well, and together they cover it.

## 2 Recovering the response curve

**WITH RAW FILES YOU MAY NOT NEED THIS AT ALL**

**Raw values are already linear in scene radiance** (Module 01), so merging exposures is a matter of dividing each by its exposure time and taking a weighted average. No curve recovery is required.

**The response curve problem exists because of JPEGs**, to which the camera has applied an unknown, proprietary, non-linear tone curve. Merging those requires inverting it first, and nobody publishes it.

**Debevec and Malik's 1997 method recovers the curve from the exposures themselves**, which was essential when raw capture was rare and is largely optional now.

**It remains worth understanding**, because the structure of the problem — recover an unknown transform from redundant, overlapping measurements — recurs constantly in computational photography, and this is its cleanest instance.

$$Z_{ij} = f(E_i \cdot \Delta t_j)$$

The value  $Z$  at pixel location  $i$  in exposure  $j$  is the camera's response  $f$  applied to the product of the scene irradiance  $E_i$  at that location and the exposure time  $\Delta t_j$ . Both  $f$  and  $E$  are unknown; only  $Z$  and  $\Delta t$  are known.

$$g(Z) \equiv \ln f^{-1}(Z) = \ln E_i + \ln \Delta t_j$$

**Taking logarithms is the whole trick.** It converts a product into a sum and makes the problem *linear in the unknowns* — the 256 values of  $g$  (one per possible pixel level) and the log-irradiance at each sampled location. With many pixels across many exposures this is massively overdetermined.

$$\min \sum w(Z) [g(Z_{ij}) - \ln E_i - \ln \Delta t_j]^2 + \lambda \sum w(z) g''(z)^2$$

A weighted least-squares problem, solved by SVD. The second term penalises curvature in  $g$ , enforcing the reasonable prior that a camera response is smooth — without it the solution is underdetermined at levels that appear rarely.

**WEIGHT BY RELIABILITY**

**Pixel values near zero are dominated by read noise** and carry little information. **Values near saturation may be clipped**, in which case they are actively misleading rather than merely noisy.

**Mid-range values are the reliable ones** — enough signal to be well above the noise floor, and far enough from saturation to be unclipped.

**So weight each measurement by a hat function** that peaks in the middle of the range and falls to zero at both ends. This is Debevec and Malik's choice and it works well.

**The principled version uses the noise model from Module 01:** weight by inverse variance. Since shot noise is  $\sqrt{N}$ , a brighter measurement has a better signal-to-noise ratio and deserves more weight — so the optimal weighting is asymmetric, favouring the brighter unclipped measurements rather than the middle.

**This matters more than it sounds** and is what modern merging implementations do.

## 3 Ghosting

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### ANYTHING THAT MOVES APPEARS SEVERAL TIMES

**Merging assumes every exposure is of the same scene.** A person who walks through the frame between the first and last exposure appears in the merged result as a semi-transparent repetition — a ghost, hence the name.

**Leaves moving in wind, water, traffic, pedestrians, and clouds are the usual culprits,** and almost every real outdoor scene contains at least one of them. Ghosting is the normal case, not the exception.

**Detection is straightforward in principle:** scale each exposure by its exposure time to bring it into a common radiance space, compare against a chosen reference, and flag regions of large disagreement that cannot be explained by noise or clipping.

**Resolution involves accepting a loss.** In a region flagged as moving, take all the data from a single exposure rather than blending — which means more noise there, or clipped highlights there, in exchange for a consistent image. **There is no way to have both,** because the different exposures genuinely recorded different scenes.

- **Reference-based selection.** Choose one exposure as the reference for structure; in regions that disagree with it, use only data consistent with that reference. Simple, robust, and the usual approach.
- **Median-based merging.** With enough exposures, taking a median rather than a mean is robust to objects appearing in a minority of frames — a person who crosses the frame once in nine exposures simply vanishes.
- **Optical flow alignment.** Estimate dense motion between exposures and warp them into correspondence, so that moving objects are merged correctly rather than excluded. **Powerful, and it fails at occlusion boundaries** where there is no correspondence to find.
- **Patch-based synthesis.** Fill the moving regions with plausible content assembled from elsewhere in the exposure stack. Effective and it is synthesis rather than measurement, which Module 13 takes up.
- **Or capture a single raw frame and avoid the problem.** With 14 stops of sensor range this is increasingly viable for ordinary scenes, and it is the honest answer for a great many cases where people still bracket out of habit.

## 4 What cameras actually do now

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**MODERN HDR DOES NOT BRACKET EXPOSURES**

**HDR+ and its successors capture a burst of *identically* underexposed raw frames** — ten to fifteen of them, all short, all deliberately dark.

**Underexposing protects the highlights**, which is the unrecoverable failure mode from §1. Nothing clips, so nothing is lost.

**Averaging the burst recovers the shadows.** Signal-to-noise improves as  $\sqrt{n}$  (Module 01), so fifteen frames gives roughly a fourfold improvement — enough to make the deliberately dark exposure usable.

**The noise that made underexposure unacceptable is averaged away**, and the highlights were never in danger.

**And identical exposures align far better than bracketed ones.** Alignment between a bright frame and a dark one is genuinely hard because the images look different; alignment between fifteen nominally identical frames is comparatively easy. So ghosting becomes tractable rather than fundamental. **This is why phone HDR works as well as it does**, and Module 10 covers the pipeline in full.

|                    | Exposure bracketing                                                                   | Underexposed burst                                               |
|--------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Frames             | 3 to 9, spanning several stops.                                                       | <b>10 to 15, nominally identical.</b>                            |
| Highlights         | Preserved only by the shortest exposure.                                              | <b>Never clipped at all.</b>                                     |
| Shadows            | From the longest exposure, which is noisy or blurred.                                 | <b>Recovered by averaging.</b>                                   |
| Alignment          | <b>Hard</b> — the frames genuinely look different, so feature matching is unreliable. | <b>Easy</b> — the frames match, so simple block alignment works. |
| Motion             | The long exposure carries motion blur.                                                | <b>Every frame is short, so no blur in any of them.</b>          |
| Total capture time | Dominated by the longest exposure — potentially a second or more.                     | <b>Short</b> , because every frame is short.                     |

**The burst approach wins on every row**, which is why it replaced bracketing in consumer cameras within a few years. Bracketing survives in landscape photography, where the scene is static and the camera is on a tripod, and as a teaching exercise — which is what Project 1 uses it for.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                                                                        | What it covers for this module                                                                               |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <a href="#">Debevec &amp; Malik — Recovering High Dynamic Range Radiance Maps from Photographs (1997, free)</a> | The paper. The log-linearisation of §2 and the weighting scheme, in the original, and still clearly written. |

| Resource                                                                               | What it covers for this module                                                                    |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Hasinoff et al. — Burst photography for HDR and low-light imaging (free, with dataset) | The §4 approach, documented by the team that shipped it, with a dataset of raw bursts to work on. |
| CMU 15-463 — HDR imaging lecture and assignment (free)                                 | The merging assignment, which is essentially Project 1's HDR requirement.                         |
| Tursun et al. — The State of the Art in HDR Deghosting (free)                          | A survey of §3's techniques, with comparisons.                                                    |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Photograph a high-contrast scene at a single exposure and demonstrate that neither highlights nor shadows can be recovered afterwards.
- 2 Capture a bracketed sequence on a tripod and recover the response curve by the Debevec–Malik method. Plot it.
- 3 Merge the same sequence from raw, where no curve recovery is needed, and compare against the JPEG-derived result.
- 4 Implement both the hat weighting and inverse-variance weighting, and compare the noise in the merged result.
- 5 Capture a bracketed sequence with a person walking through it. Demonstrate the ghosting, then implement reference-based deghosting.
- 6 Capture ten identically underexposed raw frames of a high-contrast scene. Average them and compare against a single correctly exposed frame.
- 7 Measure the noise reduction from averaging  $n$  frames for  $n = 2, 4, 8, 16$  and confirm the  $\sqrt{n}$  relationship.
- 8 Compare alignment difficulty between a bracketed set and an identical burst by running the same feature matcher on both.
- 9 Compute the dynamic range of a scene you photograph, in stops, from the merged radiance map.

## Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the dynamic range of a scene, a sensor, and a display, and say which constraint Module 04 addresses.
- 2 Why are clipped highlights unrecoverable?

- 3 What does taking logarithms achieve in the Debevec–Malik formulation?
- 4 Why might you not need response curve recovery, and why learn it anyway?
- 5 Why weight by reliability, and what is the principled weighting?
- 6 What causes ghosting, and why can it not be fully resolved?
- 7 Name four deghosting approaches.
- 8 Why does an underexposed burst beat exposure bracketing on every measure?